PF2 Worked Examples: Graphing Tangent, Cotangent, Secant, and Cosecant

Recording companion for the PF2 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Where are the vertical asymptotes? Everything else, the period, the domain, and the sketch, is built on them. Start from the parent function's asymptotes. A number multiplying x squeezes or stretches them: divide each location by that number. A number added to x shifts them: subtract it. The domain is everything except the asymptotes. The range does not change: all real numbers for tangent and cotangent, and everything at or beyond 1 and negative 1 for secant and cosecant.

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Example 1: Asymptotes 1

$$f(x)=\tan(3x)$$

Question

Where are the vertical asymptotes of this function?

- **A.** $x=\pi/2+k\pi$
- **B.** $x=\pi/6+(k\pi)/3$
- **C.** $x=(3\pi)/2+3k\pi$
- **D.** $x=(k\pi)/3$

Answer and reasoning

Tangent breaks where its input is $\pi/2+k\pi$. Here the input is $3x$, so set $3x=\pi/2+k\pi$ and divide everything by 3: $x=\pi/6+(k\pi)/3$. The 3 squeezes the graph, so the asymptotes get closer together, not farther apart. The answer $x=(k\pi)/3$ uses cotangent's asymptotes. **The asymptotes are at $x=\pi/6+(k\pi)/3$.**

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Example 2: Asymptotes 2

$$g(x)=\csc(x+\pi/4)$$

Question

Where are the vertical asymptotes of this function?

- **A.** $x=\pi/4+k\pi$
- **B.** $x=k\pi$
- **C.** $x=-\pi/4+2k\pi$
- **D.** $x=-\pi/4+k\pi$

Answer and reasoning

Cosecant breaks where its input is $k\pi$. Set $x+\pi/4=k\pi$ and subtract: $x=-\pi/4+k\pi$. Adding inside shifts the graph left, so the asymptotes move left by $\pi/4$. They stay π apart even though the period of cosecant is 2π, because one period holds two branches. **The asymptotes are at $x=-\pi/4+k\pi$.**

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Example 3: Asymptotes 3

$$h(x)=\sec(1/2x)$$

Question

Where are the vertical asymptotes of this function?

- **A.** $x=\pi+2k\pi$
- **B.** $x=\pi/4+(k\pi)/2$
- **C.** $x=\pi/2+k\pi$
- **D.** $x=2k\pi$

Answer and reasoning

Secant breaks where its input is $\pi/2+k\pi$. Set $x/2=\pi/2+k\pi$ and multiply by 2: $x=\pi+2k\pi$. Dividing by one half is the same as multiplying by 2, so the graph is stretched and the asymptotes spread out. **The asymptotes are at $x=\pi+2k\pi$.**

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Example 4: Period 1

$$f(x)=-\cot(2x)$$

Question

What is the period of this function?

A. 2π
B. $\pi/2$
C. π
D. $\pi/4$

Answer and reasoning

The period of cotangent is π. The 2 inside squeezes the graph, so divide: $\pi/2$. The minus sign in front flips the graph upside down. It does not change the period or move the asymptotes, which are at $x=(k\pi)/2$. The answer 2π multiplies by 2 instead of dividing. **The period is $\pi/2$.**

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Example 5: Domain 1

$$f(x)=\tan(2x)$$

Question

What is the domain of this function?

A. $\ldots\cup(-\pi/2,\pi/2)\cup(\pi/2,(3\pi)/2)\cup\ldots$
B. $\ldots\cup(-\pi/4,\pi/4)\cup(\pi/4,(3\pi)/4)\cup\ldots$
C. $\ldots\cup(0,\pi/2)\cup(\pi/2,\pi)\cup\ldots$
D. $(-\infty,\infty)$

Answer and reasoning

The asymptotes are where $2x=\pi/2+k\pi$, so $x=\pi/4+(k\pi)/2$: at $-\pi/4$, $\pi/4$, $(3\pi)/4$, and so on. The domain is the open intervals between them. The first answer ignores the 2. The intervals starting at 0 belong to $\cot(2x)$. All real numbers is the range, not the domain. **The domain is the open intervals between $-\pi/4$, $\pi/4$, $(3\pi)/4$, and so on.**

Example 6: Range 1

$$g(x)=\sec(x-\pi/4)$$

Question

What is the range of this function?

A. $(-\infty,\infty)$
B. $[-1,1]$
C. $(-\infty,-1]\cup[1,\infty)$
D. $(-\infty,-1)\cup(1,\infty)$

Answer and reasoning

A shift moves the graph left or right, which never changes the range. Secant is 1 divided by cosine, and cosine stays between -1 and 1, so secant is always at least 1 or at most -1. It reaches exactly 1 and -1 at the turning points of each branch, so those ends are included with square brackets. The interval $[-1,1]$ is the range of cosine. **The range is everything at or above 1 together with everything at or below negative 1.**

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Example 7: Domain 2

$$h(x)=\cot(1/3x)$$

Question

What is the domain of this function?

A. $\ldots\cup(-3\pi,0)\cup(0,3\pi)\cup\ldots$
B. $\ldots\cup(-\pi/3,0)\cup(0,\pi/3)\cup\ldots$
C. $\ldots\cup(-(3\pi)/2,(3\pi)/2)\cup((3\pi)/2,(9\pi)/2)\cup\ldots$
D. $\ldots\cup(-\pi,0)\cup(0,\pi)\cup\ldots$

Answer and reasoning

Cotangent breaks where its input is $k\pi$. Set $x/3=k\pi$ and multiply by 3: $x=3k\pi$, so the asymptotes are at -3π, 0, 3π, and so on. The intervals with $\pi/3$ divide by 3 instead of multiplying. The intervals ending at $(3\pi)/2$ use tangent's asymptotes. **The domain is the open intervals between -3π, 0, 3π, and so on.**

Example 8: Domain 3

$$f(x)=\tan(x+(3\pi)/4)$$

Question

What is the domain of this function?

A. $\ldots\cup(\pi/4,(5\pi)/4)\cup((5\pi)/4,(9\pi)/4)\cup\ldots$

B. $\ldots\cup(-\pi/2,\pi/2)\cup(\pi/2,(3\pi)/2)\cup\ldots$

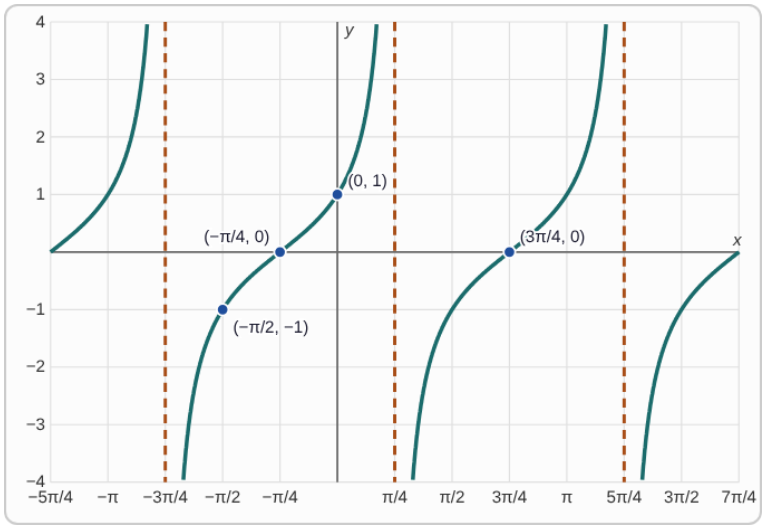
C. $\ldots\cup(-\pi/4,(3\pi)/4)\cup((3\pi)/4,(7\pi)/4)\cup\ldots$

D. $\ldots\cup(-\pi,0)\cup(0,\pi)\cup\ldots$

Answer and reasoning

Set $x+(3\pi)/4=\pi/2+k\pi$ and subtract: $x=-\pi/4+k\pi$. So the asymptotes are at $-\pi/4$, $(3\pi)/4$, $(7\pi)/4$, and so on. The intervals starting at $\pi/4$ shift the wrong way, to the right. The intervals around $\pi/2$ ignore the shift. **The domain is the open intervals between $-\pi/4$, $(3\pi)/4$, $(7\pi)/4$, and so on.**

Example 9: Graph 1



Plotted data: dashed vertical lines at x = −3π/4, π/4, and 5π/4, spaced π apart. Between each pair of neighboring dashed lines, the curve rises from the bottom of the grid to the top, getting very close to the dashed lines at both ends. It crosses the x-axis at (−π/4, 0) and (3π/4, 0), and passes through (−π/2, −1) and (0, 1). The same rising shape continues past the dashed lines at both edges.

Question

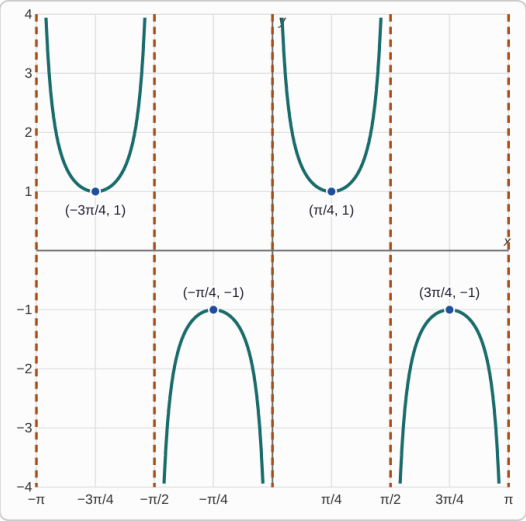
Which function has this graph?

- **A.** $f(x)=\tan(x-\pi/4)$
- **B.** $f(x)=\cot(x-\pi/4)$
- **C.** $f(x)=\cot(x+\pi/4)$
- **D.** $f(x)=\tan(x+\pi/4)$

Answer and reasoning

The curve rises between the dashed lines, which is how tangent looks; cotangent falls. For $\tan(x+\pi/4)$ the asymptotes are where $x+\pi/4=\pi/2+k\pi$, which is $x=\pi/4+k\pi$, matching the dashed lines. Check a point: $\tan(0+\pi/4)=1$, which matches $(0, 1)$. The version with $x-\pi/4$ would put the dashed lines at $(3\pi)/4$ and $-\pi/4$. **The graph is $f(x)=\tan(x+\pi/4)$.**

Example 10: Graph 2



Plotted data: dashed vertical lines at x = −π, −π/2, 0, π/2, and π, spaced π/2 apart. Between each pair of neighboring dashed lines is one U-shaped piece whose ends climb toward the dashed lines. From left to right the pieces alternate: a U opening upward with lowest point (−3π/4, 1), a U opening downward with highest point (−π/4, −1), a U opening upward with lowest point (π/4, 1), and a U opening downward with highest point (3π/4, −1). No part of the curve lies strictly between y = −1 and y = 1.

Question

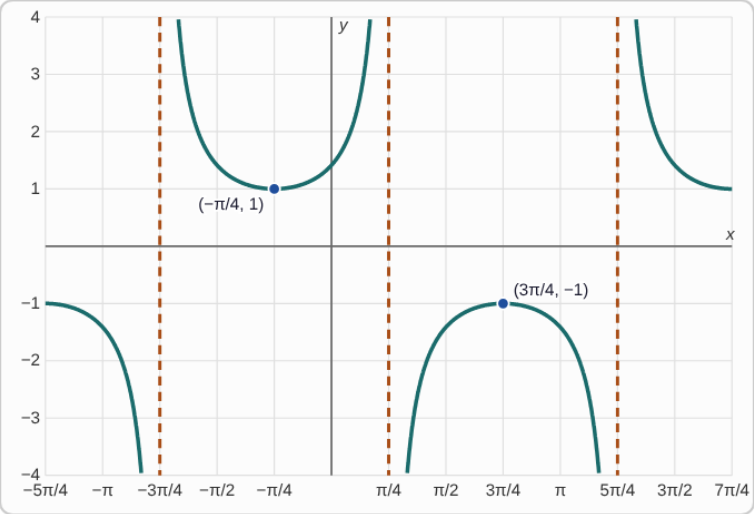
Which function has this graph?

- **A.** $f(x)=\sec(2x)$
- **B.** $f(x)=\csc(2x)$
- **C.** $f(x)=-\csc(2x)$
- **D.** $f(x)=\csc(1/2x)$

Answer and reasoning

The dashed lines are at $x=0$ and every $\pi/2$, so the asymptotes are $x=(k\pi)/2$. Cosecant has an asymptote at 0, and $2x=k\pi$ gives exactly these. Secant of $2x$ would put them at $\pi/4$, $(3\pi)/4$, and so on. Now check the direction: $\csc(2\cdot\pi/4)=\csc(\pi/2)=1$, which matches $(\pi/4, 1)$. With the minus sign in front, that point would be at -1. **The graph is $f(x)=\csc(2x)$.**

Example 11: Graph 3



Plotted data: dashed vertical lines at x = −3π/4, π/4, and 5π/4, spaced π apart. Between −3π/4 and π/4 is a U opening upward with lowest point (−π/4, 1). Between π/4 and 5π/4 is a U opening downward with highest point (3π/4, −1). Pieces of the same shapes continue past the dashed lines at both edges. No part of the curve lies strictly between y = −1 and y = 1.

Question

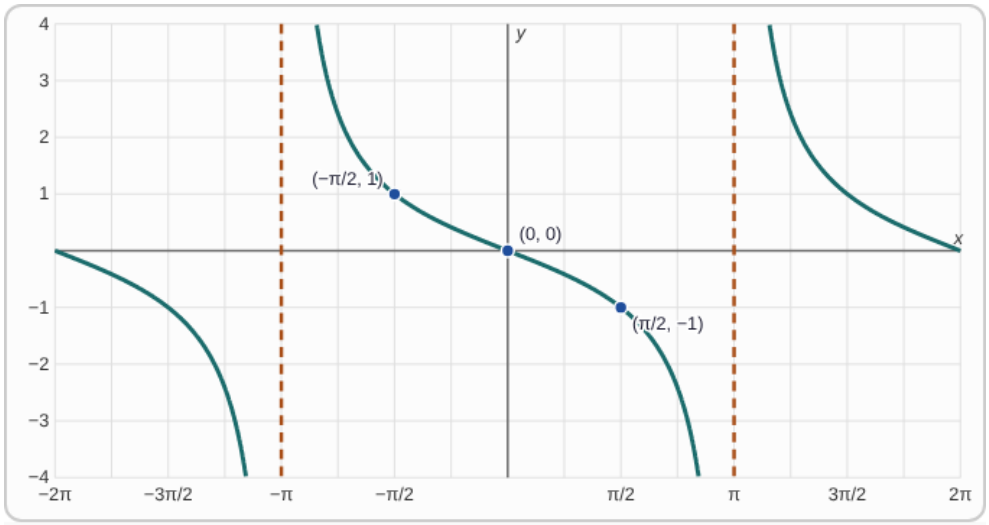
Which function has this graph?

- **A.** $f(x)=\sec(x-\pi/4)$
- **B.** $f(x)=\csc(x+\pi/4)$
- **C.** $f(x)=\sec(x+\pi/4)$
- **D.** $f(x)=\csc(x-\pi/4)$

Answer and reasoning

Two answers have dashed lines in the right places. For $\sec(x+\pi/4)$, $x+\pi/4=\pi/2+k\pi$ gives $x=\pi/4+k\pi$. For $\csc(x-\pi/4)$, $x-\pi/4=k\pi$ gives the same lines. A point decides it: $\sec(-\pi/4+\pi/4)=\sec 0=1$, which matches $(-\pi/4, 1)$, but $\csc(-\pi/2)=-1$. **The graph is $f(x)=\sec(x+\pi/4)$.**

Example 12: Graph 4



Plotted data: dashed vertical lines at x = −π and x = π, spaced 2π apart. Between them the curve falls from the top of the grid near x = −π to the bottom near x = π, passing through (−π/2, 1), (0, 0), and (π/2, −1). The same falling shape appears to the left of −π and to the right of π.

Question

Which function has this graph?

- **A.** $f(x)=\tan(1/2x)$
- **B.** $f(x)=-\tan(1/2x)$
- **C.** $f(x)=-\tan(2x)$
- **D.** $f(x)=\cot(1/2x)$

Answer and reasoning

The dashed lines are 2π apart at $\pm\pi$, and the curve crosses the x-axis at 0. For $\tan(x/2)$, $x/2=\pi/2+k\pi$ gives $x=\pi+2k\pi$, which matches. Cotangent of $x/2$ would have a dashed line at 0 instead. But the curve falls, while tangent rises, so the graph is flipped: there is a minus sign. Check: $-\tan(\pi/4)=-1$ at $x=\pi/2$. This is the same function as $\tan(-x/2)$. **The graph is $f(x)=-\tan(x/2)$.**

Example 13: Slip 1

$$f(x)=\tan(4x)$$

Question

A student says the period of this function is 4π. What is the mistake, if any?

A. Every tangent has period 2π.
B. The 4 only changes the height, so the period is still π.
C. Divide by 4, do not multiply: $\pi/4$.
D. There is no mistake.

Answer and reasoning

A number multiplying x squeezes the graph, so the period gets shorter. Tangent's period is π, so divide: $\pi/4$. A number in front of the function would change the height, but this 4 is inside. **The period is $\pi/4$.**

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Example 14: Slip 2

$$g(x)=\sec(2x)$$

Question

A student says the range of this function is $[-1,1]$. What is the mistake, if any?

A. There is no mistake.

B. The range should be all real numbers, since the graph goes up and down forever.

C. Because of the 2, the range should be $[-2,2]$.

D. That is cosine's range. Secant skips everything between negative 1 and 1.

Answer and reasoning

Secant is 1 divided by cosine. Cosine stays between -1 and 1, so its reciprocal is always at least 1 or at most -1. The 2 inside changes only the asymptotes and the period. **The range is $(-\infty,-1]\cup[1,\infty)$.**

Example 15: Slip 3

$$h(x)=\cot(x-\pi/4)$$

Question

A student says the asymptotes are at $x=-\pi/4+k\pi$. What is the mistake, if any?

A. Subtracting inside shifts right, so the asymptotes are at $x=\pi/4+k\pi$.
B. There is no mistake.
C. The spacing between asymptotes should be 2π.
D. A shift removes the asymptote at 0, so the asymptotes are at $x=\pi/2+k\pi$.

Answer and reasoning

Cotangent breaks where its input is $k\pi$. Set $x-\pi/4=k\pi$ and add: $x=\pi/4+k\pi$. The student moved the asymptotes the wrong way. Cotangent's period is π, so the spacing stays π.
The asymptotes are at $x=\pi/4+k\pi$.

Example 16: Slip 4

$$f(x)=\sec(3x)$$

Question

A student says the asymptotes are at $x=\pi/6+(k\pi)/3$, so the period is $\pi/3$. What is the mistake, if any?

A. Secant matches cosecant, so the asymptotes are at $x=(k\pi)/3$.

B. One period holds two branches, so the period is $(2\pi)/3$.

C. The 3 stretches the graph, so the period is 3π.

D. There is no mistake.

Answer and reasoning

The asymptotes are right: $3x=\pi/2+k\pi$ gives $x=\pi/6+(k\pi)/3$. But secant's period is 2π, not π: one upward branch and one downward branch make one full period. Divide by 3: $(2\pi)/3$. That is two asymptote gaps, not one. **The period is $(2\pi)/3$.**

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Example 17: Slip 5

$$f(x)=\tan(1/2x)$$

Question

A student says the asymptotes are at $x=\pi+2k\pi$, and the domain is every real number except those. What is the mistake, if any?

A. There is no mistake.
B. The asymptotes should be at $x=\pi/4+(k\pi)/2$.
C. The asymptotes should be at $x=2k\pi$.
D. The domain should be all real numbers.

Answer and reasoning

Set $x/2=\pi/2+k\pi$ and multiply by 2: $x=\pi+2k\pi$. The domain leaves out exactly those values. In CheckIt's form that is the open intervals between $-\pi$, π, 3π, and so on.
There is no mistake.